P5: Search Consolidation

FIT1061 Pass Task — Week 5 — When Algorithms Choose Who Gets Help

This task brings the earlier search ideas together. You will compare them and explain what each one does well and where it falls short.

Show us that you can

- You can compare multiple search strategies and explain their differences clearly.
- You can connect your implementation work back to the lecture ideas.
- You can reflect on the ethical and practical consequences of algorithm choice.

Give us

- A completed consolidation notebook (BFS/DFS/greedy comparison).
- A working `find_pareto_frontier` and the Pareto plot over the four targets (Part 6).
- The P5.4 ethics reflection (250–350 words), written inside the notebook.
- Your personalised hand-computation, photographed.

Resources

- Week 5 lecture slides and recording *(Moodle)*
- P5 starter notebook *(OnTrack)*
- Earlier search notes and examples *(Moodle)*

Key sections in the task notebook

- Part 1: The scenario
- Part 2: BFS, the baseline
- Part 2b: DFS, the depth-first baseline
- Part 3: Greedy best-first search, guided by a heuristic (P5.1)
- Part 4: Side-by-side comparison
- Part 5: Guided analysis questions
- Part 6: When the algorithm has to choose (graded: Pareto frontier)

Do the following

1. Review the previous search tasks before starting this consolidation task.
2. Work through the notebook and compare the algorithms directly.
3. Write your explanation in your own words rather than copying the lecture wording.
4. Upload your completed work and mark it ready for feedback.

! Acceptable use of Artificial Intelligence

AI tools are **discouraged** for implementation tasks at the Pass level. You may use AI for *explanation* (e.g., “explain what a queue is”), but not to generate code or write your reflection. Cite all AI use. The

milestone discussion will probe for genuine understanding.
All submitted work must be your own.